+49 178 444 73 43
atamererkal.eu@gmail.com
linkedin.com/in/atamererkal
github.com/AtamerErkal

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Scan QR Code
or click QR Code

Atamer Erkal

AI Engineer | MLOps
Healthcare & Defence AI

Former NATO Planning Officer | EU Work Permit
Southern Germany (Open to Hybrid/Remote)

PROFIL

*AI Engineer with 15+ years of experience in **mission-critical and regulated environments**, including NATO operations.*
*Specialized in **production-grade Machine Learning and MLOps**, with hands-on experience in building, deploying, and monitoring end-to-end ML pipelines using **Azure ML, MLflow, DVC, and Docker**.*
Strong focus on:

- *Real-time and edge AI systems*
- *Azure-based ML systems (Azure ML, deployment, monitoring)*
- *GDPR & EU AI Act compliance for high-risk AI systems*
- *Explainable AI (SHAP, LIME, Grad-CAM)*

Proven ability to translate complex operational requirements into scalable, production-grade Machine Learning, MLOps, and Generative AI systems (Azure OpenAI, RAG), and compliant AI solutions on cloud and edge environments.

CORE SKILLS

Machine Learning & AI

- Supervised & Unsupervised Learning (Classification, Regression, Clustering)
- Deep Learning (CNNs, EfficientNet, YOLOv8 – PyTorch, TensorFlow)
- Computer Vision, Medical Imaging, NLP, Generative AI (RAG, Azure OpenAI, LLMs)

MLOps & Cloud

- End-to-end ML pipelines (MLflow, DVC), CI/CD for ML (GitHub Actions)
- Containerization (Docker)
- Azure ML (training, deployment, monitoring)
- REST APIs (FastAPI)
- Model monitoring, drift detection, retraining strategies

Data & Tools

- Python (pandas, NumPy, scikit-learn)
- SQL (PostgreSQL)
- Git

Domain Expertise

- Healthcare AI (FHIR/HL7, clinical NLP, imaging)
- Defence AI (real-time systems, NATO-compatible architectures)
- GDPR & EU AI Act compliance

PROFESSIONAL EXPERIENCE

Machine Learning Engineer & MLOps Lead (Program Director)
Clarusway – USA & Germany | *04/2020 – present*

- *Designed and deployed end-to-end ML pipelines using MLflow, DVC, and Azure ML*
- *Built production-ready AI systems with focus on scalability, monitoring, and compliance*
- *Developed Explainable AI systems for high-risk applications in healthcare and defence*
- *Implemented CI/CD pipelines for ML workflows using GitHub Actions and Docker*
- *Led multiple ML projects from data ingestion to deployment and monitoring*
- *Mentored 100+ data professionals; contributed to ~90% job placement rate*
- *Collaborated with international stakeholders to deliver GDPR- and EU AI Act-compliant solutions*
- ***Award:** Best Data Science & Analytics Program (2021–2022).*

Transition Phase and Full-time Upskilling in Data Science (Further Training etc.) | *03/2017 – 03/2020*

Planning Officer & Electrical Engineer – Safety-Critical Systems
NATO & Naval Headquarters – United Kingdom & Turkey | *09/2002 – 02/2017*

- *Worked in **highly regulated, multinational, mission-critical environments***
- *Performed data-driven analysis for operational planning and decision support*

- *Applied quantitative methods under **time-critical and security constraints***
- *Designed structured analytical approaches for performance evaluation and resource allocation*
- ***Award:** Certificate of Honour for Leadership and Management (Naval Forces Commander)*

SELECTED PROJECTS

(Further projects and live demos available on my portfolio: ***atamererkal.github.io*** – Scan the QR code or visit directly)

- ***VitalNav.ai – AI-Powered Pre-Hospital Triage System (Healthcare - Flagship)***
  - *Built a pre-hospital triage system using Azure OpenAI (LLMs), Retrieval-Augmented Generation (RAG), and Azure AI Speech Services*
  - *Achieved <5 seconds response time for emergency classification*
  - *Designed multilingual pipeline with GDPR-compliant architecture*
- ***AEGIS-Cloud – Autonomous AI Edge Guardian (Defence - Flagship)***
  - *Developed YOLOv8-based object detection system for edge devices*
  - *Optimized inference using TensorRT (<30 ms latency)*
  - *Implemented full pipeline including deployment and monitoring*
- ***SkinxAI – Explainable Medical Image Classification (Healthcare & Explainable AI)***
  - *CNN-based classification (EfficientNet)*
  - *Applied Grad-CAM for explainability*
  - *Achieved 85% sensitivity for malignant detection*
- ***VANGUARD – Air Track Classification (NATO Context)***
  - *Built classification model (XGBoost) with 96%+ accuracy*
  - *Designed system aligned with NATO-compatible architecture.*
- ***Explainable AI Dashboard (Responsible AI)***
  - *Built interactive dashboard for model interpretation using SHAP &LIME*
  - *Support transparency and traceability for high-risk AI systems according to the EU AI Act.*

VOLUNTEER WORK

German Red Cross (DRK) & Support Facilities

- *Support in multilingual communication and daily organization.*

EDUCATION

- **M.A. Management & Leadership**
  National Defence University, Istanbul | 2010–2012
  Award: Winner of the university-wide article competition (Thesis / Management Essay)
- **B.Sc. Electrical Engineering & Officer Training**
  National Defence University, Istanbul | 1998–2002

CERTIFICATES & FURTHER TRAINING

Further Training:

- *DataJarLabs: Data Science and Analytics | 2018–2019*
- *NATO School (Germany): Comprehensive Operations Planning*
- *Ankara University: International Relations (Further Qualification)*

Certificates:

- *Microsoft AI Engineer Associate (AI-102) – scheduled May 2026*
- *Microsoft: AI on Azure, Azure ML, Power BI & Excel for Data Analysis*
- *DataCamp: Data Scientist with Python, MLOps, CI/CD for ML*
- *Agile with Jira (Atlassian)*
- *Power BI & Excel für Datenanalyse*

Regulatory Further Training (ongoing):

- *GDPR for Data Science & AI*
- *EU AI Act – Risk Classification & Compliance*

LANGUAGES

Turkish (Native), English (Fluent), German (B2 – TELC, Exam 04/2025)

REFERENCES

Available upon request (including Clarusway, NATO contacts)